

Research Answers

Question 1

Conduct research to understand how HTTP applications preserve the state of an application across multiple request-response cycles, especially concerning user authentication and session management.

HTTP is a stateless protocol, which means the server does not remember previous requests from a user. Each request between the client and server is treated as a separate interaction. Because of this, web applications need a way to remember users between requests.

One common way to solve this problem is by using cookies. Cookies are small pieces of data stored in the user's browser. When a user logs into a website, the server creates a session and sends a session ID to the browser in the form of a cookie. When the user makes another request, the browser sends the cookie back to the server, allowing the server to identify the user and maintain their login session.

Frameworks such as Django provide built-in session management systems. Django stores session information on the server, usually in a database, cache, or file system. The browser only stores the session ID, which improves security. By using cookies and sessions, web applications can remember users and maintain authentication even though HTTP itself is stateless.

Question 2.

Investigate and document the procedures for performing Django database migrations to a server-based relational database like MariaDB.

Django migrations are used to manage changes to the database structure over time. When working with a relational database such as MariaDB, migrations allow developers to create and update database tables based on the models defined in the Django project.

The first step is to configure the database connection in the settings.py file. To connect Django with MariaDB, a database driver such as mysqlclient or PyMySQL must be installed. The database configuration then includes the database name, username, password, host, and port.

After the connection is set up, migrations can be performed using Django management commands. The command `python manage.py makemigrations` creates migration files that describe changes made to the models. These files are stored in the migrations folder of the app.

The command `python manage.py migrate` then applies those changes to the MariaDB database by creating or modifying the necessary tables.

This process helps keep the database structure organised and consistent across development and production environments.